

Aman Behera

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EDUCATION

Indian Institute of Technology, Roorkee
Bachelors in Technology — Chemical Engineering

Oct 2022 – May 2026
CGPA: 8.624 / 10

EXPERIENCE

- Data Scientist PPO | HiLabs Inc** Feb 2026 - Present
- Implemented a high-performance entity resolution pipeline using DSU (Union-Find) with blocking & reducing duplicate detection from $O(n^2)$ to block-bounded comparisons & achieving $\sim 10\times$ runtime speedup on tabular data
 - Built and deployed a containerized data analytics system with FastAPI, MySQL, and a locally hosted Small Language Model (SLM) for Natural language to SQL
- ML Systems Intern | Infoedge India Ltd** May 2025 – Jul 2025
- Worked on Generative Engine Optimization (GEO) to improve content visibility in LLM-powered search by fine-tuning Qwen3-8B-Instruct using SFT and DPO for higher fluency, credibility, and citation quality.
 - Designed objective metrics (position-adjusted word count, sentence position) and GEval-GPT based subjective evaluation. Built workflow with rewriting agents & evaluation modules for testing strategies.
- Google Summer of Code 2026 Contributor | GNU Octave** May 2026 – Aug 2026
- Selected for GSoC 2026 to design and implement an object-oriented ANOVA framework for GNU Octave
 - Built a classdef-based ANOVA framework with lazy refitting and backend routing, unifying `anova1/2/anovan` and reducing $O(N\times G)$ to $O(N)$ via vectorization
 - 25+ merged PRs to [GNU Octave](#) improving KD-Tree, kNN, HNSW, `pdist/pdist2`, median performance & memory efficiency. Key PRs: Octave Statistics Library: [#361](#), [#357](#), [#360](#), [#371](#), [#374](#), [#378](#), [#384](#), Core octave: [#40](#)

PROJECTS

- ZeroShred: P2P File Transfer System** Sep 2025 – Jan 2026
- Built a high-performance P2P file transfer system in C++ using zero-copy I/O primitives to stream data directly from disk to network, achieving up to 780 MB/s throughput while minimizing user-kernel data movement.
 - Performed SHA-256 integrity verification in a parallel thread, decoupled from the data transfer path.
- Distributed Job Queue System** Oct 2025 – Dec 2025
- Designed and implemented a distributed job queue using FastAPI, Redis, and MySQL with worker-based execution, supporting task retries, idempotency, and horizontal scaling.
 - Built rate-limiting, backpressure handling, and failure recovery mechanisms; benchmarked throughput and latency under concurrent workloads.

ACADEMIC ACHIEVEMENTS & COMPETITIONS

- Secured **AIR 4759** in JEE Advanced 2022 among 1.5 lakh candidates
- Silver Medal**, Inter IIT Tech Meet 2025, representing IIT Roorkee
- Winner**, AIQuest Hackathon 2025 conducted by **HiLabs Inc**
- Ranked **67th** (Task 5) and **24th** (Task 2) in **highload.fun** performance engineering challenges

OLYMPIAD EXPERIENCE

- Awarded with **NTSE Scholarship**, 2020 by NCERT
- Kishore Vaigyanik Protsahan Yojana (KVPY)**, 2021, **AIR 122** in SA Stream
- Indian Olympiad Qualifier in Chemistry (IOQC)**, 2021, **Top 300** students nationwide

TECHNICAL SKILLS

Programming: C++, Python, SQL, Bash
Backend: FastAPI, REST APIs, MySQL
Systems: OS Internals, Memory Management, Networking, Performance Optimization
ML Systems: PyTorch, NumPy, FAISS

POSITIONS OF RESPONSIBILITY

- Project Head, ACM Student Chapter | IIT Roorkee** Apr 2025 – Present
- Leading **50+** members; organizing talks, workshops, and projects in core CS and ML
- Member, Vision And Language Group (VLG) | IIT Roorkee** Jan 2024 – Present
- Participating in paper discussions, workshops, and projects focused on Deep learning, NLP, and CV
- Competitive Programmer | Codeforces, LeetCode** Aug 2024 – Present
- Codeforces max rating **1743** (Expert), handle: `heWhoCooks`; LeetCode rating **1704**, handle: `beingamanforever`